Chromatographic Method Development and Validation

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Chromatographic Method Development and Validation (A07825)

Short Title: Chroma Method Dev & Validation
Faculty Science and Computing
Department Science
Cluster Analytical Science
Author DCONNOLLY
Credits 5 **Level:** Intermediate

Description of Module / Aims

This module provides an outline of method planning and development in the analytical process. The importance and practices of validation activities and analytical method transfer will be described. This module also expands on the theory of chromatographic separation techniques to give an advanced understanding of the use of High Performance Liquid Chromatography (HPLC) and Gas Chromatography (GC) with respect to pharmaceutical, food and environmental applications. A necessary background will be provided to give the student the ability to select an appropriate separation technique and develop a method for a given analysis. The principles and practices of the validation of chromatographic methods will also be emphasised.

Programmes

CHRO-0001	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	3 / 5 / M
CHRO-0001	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	3 / 5 / M
CHRO-0001	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	3 / 5 / M
CHRO-0001	BSc in Pharmaceutical Science (WD_SPHSC_D)	3 / 5 / M

Indicative Content

- Optimisation and the quantitative evaluation of liquid-liquid extraction methods
- Principles of chromatographic band broadening and strategies used to minimise band broadening in both HPLC and GC methods
- Quantitative evaluation of chromatographic performance
- Design of HPLC columns (particle technology and stationary phase chemistry)
- Method development strategies for neutral and ionisable molecules in reversed phase liquid chromatography
- Development of gradient methods in liquid chromatography
- Qualitative and quantitative analysis in liquid chromatography
- Method validation strategies
- Sample preparation strategies prior to chromatography, including filtration, centrifugation, ultrafiltration, preconcentration
- Associated laboratory-based experiments to supplement and augment the above content

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Determine the keys factors associated with method development and relate how method parameters will affect subsequent chromatography.
2. Demonstrate the principles and practices of analytical method validation and outline a systematic approach to validation experimentation.
3. Apply High Performance Liquid Chromatography (HPLC) and Gas Chromatography (GC) to the separation of analytes.
4. Compare and contrast the techniques, instrumentation, apparatus and sample preparation techniques for HPLC and GC.
5. Operate, calibrate and qualify associated laboratory instrumentation for sample analysis.
6. Employ a variety of sample preparation procedures to pre-treat a sample prior to chromatographic analysis.

Learning and Teaching Methods

- Lectures.
- Practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,6
Continuous Assessment	50%	
Lab Report	50%	5

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of the learning outcomes. Lack of competence with regards to use of associated laboratory skills, techniques and instrumentation.

40%-49%: Will be able to show limited understanding of the fundamentals of the learning outcomes. Will display some ability with regard to the use of associated laboratory skills, techniques and instrumentation.

50%-59%: As well as the above, the student will be to show reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to use of associated laboratory techniques instrumentation and good laboratory practice.

60%-69%: As well as the above, the student will be to show reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to use of associated laboratory techniques instrumentation and good laboratory practice.

70%-100%: All of the above to excellent levels. In addition the learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to use of associated laboratory techniques and instrumentation and excellent laboratory practice.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	36	
Independent Learning	75	

Essential Material

- Harris, D.C. _Quantitative Chemical Analysis_. 8th ed.. New York: W.H. Freeman and Company, 2010.
- Snyder, L.R., J.J. Kirkland and J.W. Dolan. _Introduction to Modern liquid chromatography_. 3rd ed.. New Jersey, USA: Wiley and Sons, 2010.

Supplementary Material

- "LC-GC Chromatography Online." http://www.chromatographyonline.com/

Requested Resources

- Room Type: Chemistry Lab